symcoder round-trip demo

electrons=(a,b), muons=(p)

1 Setup

Particle types: electrons = $\{a, b\}$, muons = $\{p\}$
Group G: $G = S_{electrons} \times S_{muons} = S_2 \times S_1$, order = 2

Operations:
$|\mathbf{x}|$ (rank 1, symmetric)
$\mathbf{x} \cdot \mathbf{y}$ (rank 2, symmetric)
$\mathbf{x}_T \cdot \mathbf{y}_T$ (rank 2, symmetric)
$\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (rank 3, antisymmetric)

Event vectors (3-D):
a: $\mathbf{a} = (1.20,\ 0.30,\ 0.50)$

b: $\mathbf{b} = (0.40,\ 1.10,\ 0.20)$

c: $\mathbf{c} = (-0.50,\ 1.20,\ -0.70)$

p: $\mathbf{p} = (0.00,\ 0.00,\ 1.00)$

q: $\mathbf{q} = (0.00,\ 0.00,\ -1.00)$

2 Phase 1 Encoding (one orbit per FlavouredOperator in repS)

7 FlavouredOperators in repS

2.1 FlavouredOperator: $|\mathbf{x}|$ flavour $= (1, 0)$

Canonical representative: $|\mathbf{a}|$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $|\mathbf{a}|$, $|\mathbf{b}|$
Eval values: $+1.3342$, $+1.1874$
Encoded [2 reals]: $+1.1874$, $+1.3342$

2.2 FlavouredOperator: $|\mathbf{x}|$ flavour $= (0, 1)$

Canonical representative: $|\mathbf{p}|$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $|\mathbf{p}|$

Eval values: $+1.0000$
Encoded [1 reals]: $+1.0000$

2.3 FlavouredOperator: $\mathbf{x} \cdot \mathbf{y}$ flavour $= (2, 0)$

Canonical representative: $\mathbf{a} \cdot \mathbf{b}$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $\mathbf{a} \cdot \mathbf{b}$
Eval values: $+0.9100$
Encoded [1 reals]: $+0.9100$

2.4 FlavouredOperator: $\mathbf{x} \cdot \mathbf{y}$ flavour $= (1, 1)$

Canonical representative: $\mathbf{a} \cdot \mathbf{p}$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $\mathbf{a} \cdot \mathbf{p}$, $\mathbf{b} \cdot \mathbf{p}$
Eval values: $+0.5000$, $+0.2000$
Encoded [2 reals]: $+0.2000$, $+0.5000$

2.5 FlavouredOperator: $\mathbf{x}_T \cdot \mathbf{y}_T$ flavour $= (2, 0)$

Canonical representative: $\mathbf{a}_T \cdot \mathbf{b}_T$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $\mathbf{a}_T \cdot \mathbf{b}_T$
Eval values: $+0.8100$
Encoded [1 reals]: $+0.8100$

2.6 FlavouredOperator: $\mathbf{x}_T \cdot \mathbf{y}_T$ flavour $= (1, 1)$

Canonical representative: $\mathbf{a}_T \cdot \mathbf{p}_T$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $\mathbf{a}_T \cdot \mathbf{p}_T$, $\mathbf{b}_T \cdot \mathbf{p}_T$
Eval values: $+0.0000$, $+0.0000$
Encoded [2 reals]: $+0.0000$, $+0.0000$

2.7 FlavouredOperator: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ flavour $= (2, 1)$

Canonical representative: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$
Encoder: `HalfSortEncoder` — stores $|\text{eval}|$ of sign$= +1$ representatives

Orbit atoms: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$
Eval values: $+1.2000$
Encoded [1 reals]: $+1.2000$

3 Phase 2 Encoding (OverlapBlocks, complementarity-drop disabled)

3.1 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(0, 1)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(|\mathbf{p}|,\ |\mathbf{p}|) \to (+1.0000,\ +1.0000)$

3.2 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +2.0000, +2.5216, -0.5842, +2.5216
Decoded pairs (u, v):
$(|\mathbf{p}|,\ |\mathbf{a}|) \to (+1.0000,\ +1.3342)$
$(|\mathbf{p}|,\ |\mathbf{b}|) \to (+1.0000,\ +1.1874)$

3.3 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +2.0000, +0.0000, +1.0000, +0.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \mathbf{a}_T \cdot \mathbf{p}_T) \to (+1.0000,\ +0.0000)$
$(|\mathbf{p}|,\ \mathbf{b}_T \cdot \mathbf{p}_T) \to (+1.0000,\ +0.0000)$

3.4 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +2.0000, +0.7000, +0.9000, +0.7000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \mathbf{a} \cdot \mathbf{p}) \to (+1.0000,\ +0.5000)$
$(|\mathbf{p}|,\ \mathbf{b} \cdot \mathbf{p}) \to (+1.0000,\ +0.2000)$

3.5 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=2 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 2
Encoded values: +1.0000, +0.8100
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (+1.0000,\ +0.8100)$

3.6 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=2 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 2
Encoded values: +1.0000, +0.9100
Decoded pairs (u, v):

$(|\mathbf{p}|,\ \mathbf{a}\cdot\mathbf{b}) \rightarrow (+1.0000,\ +0.9100)$

3.7 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=2 **PairFlavour** overlap= $(0,1)$, type=`12`, kind=`ASSOC`, encoded dim= 2
Encoded values: +2.0000, +2.4400
Decoded pairs (u,v):
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p})) \rightarrow (+1.0000,\ +1.2000)$
$(|\mathbf{p}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p})) \rightarrow (+1.0000,\ -1.2000)$

3.8 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=2 **PairFlavour** overlap= $(0,0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 2
Encoded values: +3.5661, +3.1900
Decoded pairs (u,v):
$(|\mathbf{a}|,\ |\mathbf{b}|) \rightarrow (+1.1874,\ +1.3342)$
$(|\mathbf{b}|,\ |\mathbf{a}|) \rightarrow (+1.3342,\ +1.1874)$
PairFlavour overlap=(1, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(1,0)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u,v):
$(|\mathbf{a}|,\ |\mathbf{a}|) \rightarrow (+1.1874,\ +1.1874)$
$(|\mathbf{b}|,\ |\mathbf{b}|) \rightarrow (+1.3342,\ +1.3342)$

3.9 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(0,0)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +2.5216, +0.0000, +1.5842, +0.0000
Decoded pairs (u,v):
$(|\mathbf{a}|,\ \mathbf{b}_T\cdot\mathbf{p}_T) \rightarrow (+1.3342,\ +0.0000)$
$(|\mathbf{b}|,\ \mathbf{a}_T\cdot\mathbf{p}_T) \rightarrow (+1.1874,\ +0.0000)$
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(1,0)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +2.5216, +0.0000, +1.5842, +0.0000
Decoded pairs (u,v):
$(|\mathbf{a}|,\ \mathbf{a}_T\cdot\mathbf{p}_T) \rightarrow (+1.3342,\ +0.0000)$
$(|\mathbf{b}|,\ \mathbf{b}_T\cdot\mathbf{p}_T) \rightarrow (+1.1874,\ +0.0000)$

3.10 Block: $|\mathbf{x}| \times \mathbf{x}\cdot\mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(0,0)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +2.5216, +0.7000, +1.4842, +0.9046
Decoded pairs (u,v):
$(|\mathbf{a}|,\ \mathbf{b}\cdot\mathbf{p}) \rightarrow (+1.3342,\ +0.2000)$
$(|\mathbf{b}|,\ \mathbf{a}\cdot\mathbf{p}) \rightarrow (+1.1874,\ +0.5000)$
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(1,0)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +2.5216, +0.7000, +1.4842, +0.8606

Decoded pairs (u, v):
$(|\mathbf{a}|,\ \mathbf{a}\cdot\mathbf{p})\rightarrow(+1.3342,\ +0.5000)$
$(|\mathbf{b}|,\ \mathbf{b}\cdot\mathbf{p})\rightarrow(+1.1874,\ +0.2000)$

3.11 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +2.5216, +1.6200, +0.9281, +2.0425
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \mathbf{a}_T\cdot\mathbf{b}_T)\rightarrow(+1.3342,\ +0.8100)$
$(|\mathbf{b}|,\ \mathbf{a}_T\cdot\mathbf{b}_T)\rightarrow(+1.1874,\ +0.8100)$

3.12 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +2.5216, +1.8200, +0.7561, +2.2947
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \mathbf{a}\cdot\mathbf{b})\rightarrow(+1.3342,\ +0.9100)$
$(|\mathbf{b}|,\ \mathbf{a}\cdot\mathbf{b})\rightarrow(+1.1874,\ +0.9100)$

3.13 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +2.5216, +0.0000, +3.0242, -0.1761
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+1.3342,\ +1.2000)$
$(|\mathbf{b}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+1.1874,\ -1.2000)$

3.14 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 1) type=11_sigma kind=ASSOC dim=2 **PairFlavour** overlap= $(0, 1)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 2
Encoded values: +0.0000, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T\cdot\mathbf{p}_T,\ \mathbf{b}_T\cdot\mathbf{p}_T)\rightarrow(+0.0000,\ +0.0000)$
$(\mathbf{b}_T\cdot\mathbf{p}_T,\ \mathbf{a}_T\cdot\mathbf{p}_T)\rightarrow(+0.0000,\ +0.0000)$
PairFlavour overlap=(1, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(1, 1)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(\mathbf{a}_T\cdot\mathbf{p}_T,\ \mathbf{a}_T\cdot\mathbf{p}_T)\rightarrow(+0.0000,\ +0.0000)$
$(\mathbf{b}_T\cdot\mathbf{p}_T,\ \mathbf{b}_T\cdot\mathbf{p}_T)\rightarrow(+0.0000,\ +0.0000)$

3.15 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +0.0000, +0.7000, -0.1000, +0.0000

Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{b} \cdot \mathbf{p})\to (+0.0000,\ +0.5000)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{p})\to (+0.0000,\ +0.2000)$
PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(1, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +0.0000, +0.7000, -0.1000, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{p})\to (+0.0000,\ +0.5000)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{b} \cdot \mathbf{p})\to (+0.0000,\ +0.2000)$

3.16 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +0.0000, +1.6200, -0.6561, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T)\to (+0.0000,\ +0.8100)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T)\to (+0.0000,\ +0.8100)$

3.17 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +0.0000, +1.8200, -0.8281, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{b})\to (+0.0000,\ +0.9100)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{b})\to (+0.0000,\ +0.9100)$

3.18 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(1, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +0.0000, +0.0000, +1.4400, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))\to (+0.0000,\ -1.2000)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))\to (+0.0000,\ +1.2000)$

3.19 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11_sigma kind=ASSOC dim=2 **PairFlavour** overlap= $(0, 1)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 2
Encoded values: +0.9899, +0.2900
Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{b} \cdot \mathbf{p})\to (+0.2000,\ +0.5000)$
$(\mathbf{b} \cdot \mathbf{p},\ \mathbf{a} \cdot \mathbf{p})\to (+0.5000,\ +0.2000)$
PairFlavour overlap=(1, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(1, 1)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{a} \cdot \mathbf{p})\to (+0.2000,\ +0.2000)$
$(\mathbf{b} \cdot \mathbf{p},\ \mathbf{b} \cdot \mathbf{p})\to (+0.5000,\ +0.5000)$

3.20 Block: $\mathbf{x}\cdot\mathbf{y}\times\mathbf{x}_T\cdot\mathbf{y}_T$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(1,0)$, type=`11`, kind=`ASSOC`, encoded dim= 4

Encoded values: +0.7000, +1.6200, -0.5561, +0.5670

Decoded pairs (u, v):

$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{a}_T\cdot\mathbf{b}_T)\to$ (+0.5000, +0.8100)

$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{a}_T\cdot\mathbf{b}_T)\to$ (+0.2000, +0.8100)

3.21 Block: $\mathbf{x}\cdot\mathbf{y}\times\mathbf{x}\cdot\mathbf{y}$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(1,0)$, type=`11`, kind=`ASSOC`, encoded dim= 4

Encoded values: +0.7000, +1.8200, -0.7281, +0.6370

Decoded pairs (u, v):

$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{a}\cdot\mathbf{b})\to$ (+0.5000, +0.9100)

$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{a}\cdot\mathbf{b})\to$ (+0.2000, +0.9100)

3.22 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(1,1)$, type=`11`, kind=`ASSOC`, encoded dim= 4

Encoded values: +0.7000, +0.0000, +1.5400, -0.3600

Decoded pairs (u, v):

$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to$ (+0.5000, +1.2000)

$(\mathbf{b}\cdot\mathbf{p},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to$ (+0.2000, -1.2000)

3.23 Block: $\mathbf{x}_T\cdot\mathbf{y}_T\times\mathbf{x}_T\cdot\mathbf{y}_T$ (block)

PairFlavour overlap=(2, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(2,0)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0

Decoded pairs (u, v):

$(\mathbf{a}_T\cdot\mathbf{b}_T,\ \mathbf{a}_T\cdot\mathbf{b}_T)\to$ (+0.8100, +0.8100)

3.24 Block: $\mathbf{x}_T\cdot\mathbf{y}_T\times\mathbf{x}\cdot\mathbf{y}$ (block)

PairFlavour overlap=(2, 0) type=11 kind=ASSOC dim=2 **PairFlavour** overlap= $(2,0)$, type=`11`, kind=`ASSOC`, encoded dim= 2

Encoded values: +0.8100, +0.9100

Decoded pairs (u, v):

$(\mathbf{a}_T\cdot\mathbf{b}_T,\ \mathbf{a}\cdot\mathbf{b})\to$ (+0.8100, +0.9100)

3.25 Block: $\mathbf{x}_T\cdot\mathbf{y}_T\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=2 **PairFlavour** overlap= $(2,0)$, type=`12`, kind=`ASSOC`, encoded dim= 2

Encoded values: +1.6200, +2.0961

Decoded pairs (u, v):

$(\mathbf{a}_T\cdot\mathbf{b}_T,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to$ (+0.8100, +1.2000)

$(\mathbf{a}_T\cdot\mathbf{b}_T,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to$ (+0.8100, -1.2000)

3.26 Block: $\mathbf{x}\cdot\mathbf{y}\times\mathbf{x}\cdot\mathbf{y}$ (block)

PairFlavour overlap=(2, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(2,0)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(\mathbf{a}\cdot\mathbf{b},\ \mathbf{a}\cdot\mathbf{b})\rightarrow(+0.9100,\ +0.9100)$

3.27 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=2 **PairFlavour** overlap= $(2,0)$, type=`12`, kind=`ASSOC`, encoded dim= 2
Encoded values: +1.8200, +2.2681
Decoded pairs (u, v):
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+0.9100,\ +1.2000)$
$(\mathbf{a}\cdot\mathbf{b},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+0.9100,\ -1.2000)$

3.28 Block: $\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(2, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(2,1)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+1.2000,\ +1.2000)$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(-1.2000,\ -1.2000)$

4 Alignment Decoder (recovers the G-orbit of repS evaluations)

—**repS**—: $|repS| = 7$

—**G**—: $|G| = 2$

Decoded vectors: decoded column vectors: 2

Decoded alignment table (rows=repS atoms, cols=group elements): **Decoded alignment table** (rows= repS atoms, columns= $g \in G$):

Table 1: Decoded alignment table

Atom	g_0	g_1
$\lvert\mathbf{a}\rvert$	+1.1874	+1.3342
$\lvert\mathbf{p}\rvert$	+1.0000	+1.0000
$\mathbf{a}\cdot\mathbf{b}$	+0.9100	+0.9100
$\mathbf{a}\cdot\mathbf{p}$	+0.2000	+0.5000
$\mathbf{a}_T\cdot\mathbf{b}_T$	+0.8100	+0.8100
$\mathbf{a}_T\cdot\mathbf{p}_T$	+0.0000	+0.0000
$\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p})$	-1.2000	+1.2000

Ground truth (direct evaluation of g·repS on event): **Ground truth** (direct evaluation of $g\cdot$repS on event):

Table 2: Ground truth alignment table

Atom	g_0	g_1
$\lvert\mathbf{a}\rvert$	+1.3342	+1.1874
$\lvert\mathbf{p}\rvert$	+1.0000	+1.0000
$\mathbf{a}\cdot\mathbf{b}$	+0.9100	+0.9100
$\mathbf{a}\cdot\mathbf{p}$	+0.5000	+0.2000
$\mathbf{a}_T\cdot\mathbf{b}_T$	+0.8100	+0.8100
$\mathbf{a}_T\cdot\mathbf{p}_T$	+0.0000	+0.0000
$\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p})$	+1.2000	-1.2000

Multiset comparison (decoded vs ground truth): **Multiset comparison: MATCH** (multisets equal, atol= 10^{-8})

Max —error— per repS row (decoded vs ground truth): **Max |error| per repS row:**
Mymag(a): $|\mathbf{a}|$: max err $= 0.00e + 00$

Mymag(p): $|\mathbf{p}|$: max err $= 0.00e + 00$

dot(a,b): $\mathbf{a}\cdot\mathbf{b}$: max err $= 0.00e + 00$

dot(a,p): $\mathbf{a}\cdot\mathbf{p}$: max err $= 0.00e + 00$

dot(a,b): $\mathbf{a}_T\cdot\mathbf{b}_T$: max err $= 0.00e + 00$

dot(a,p): $\mathbf{a}_T\cdot\mathbf{p}_T$: max err $= 0.00e + 00$

eps(a,b,p): $\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p})$: max err $= 0.00e + 00$

5 Summary

Phase 1 encoded length: Phase 1 encoded length: 10 reals

Phase 2 encoded length: Phase 2 encoded length: 90 reals

Total encoded length: Total encoded length: 100 reals

repS size: |repS| = 7 atoms

—**G**—: $|G| = 2$ group elements

Decoded vectors: decoded alignment vectors: 2

TeX output written to: /Users/lester/github/Echinocoder/symcoder/demo_roundtrip.tex Compile with: pdflatex demo_roundtrip.tex